G M Tasnim Alam

+1-631-590-9868 | shovon54@gmail.com | shovon0203.github.io | LinkedIn | GitHub | Stony Brook, NY

Ph.D. candidate; former Apple SWE intern and Agoda senior SWE.

Education

State University of New York at Stony Brook Aug 2023 – Expected 2028
Ph.D. in Computer Science, Advisor: Prof. Omar Chowdhury *Stony Brook, NY*

- **Research:** stateful protocol fuzzing, automated vulnerability discovery, ASN.1 / X.509 parsing and validation, 5G core network security.

Research Experience

Graduate Research Assistant, Reasoners Lab Aug 2023 – Present
Stony Brook University *Stony Brook, NY*

- **Stateful 5G core fuzzing:** Found **45 vulnerabilities (26 CVEs**: 1 Critical CVSS 9.8, 5 High, 20 Medium) in widely deployed open-source 5G cores by designing and building a **stateful black-box fuzzer** that mutates NGAP control-plane messages across multi-step sessions, reaching protocol states stateless fuzzers cannot. First-author paper under submission.
- Triaged and root-caused memory-safety and state-handling bugs in large **C/C++** codebases (OpenAirInterface, Linux Foundation projects) and drove coordinated fixes with upstream maintainers.
- **X.509 certificate validation:** Building a **grey-box fuzzing framework** for certificate chain validation libraries; libraries tested GNUTLS, MatrixSSL, WolfSSL.
- **eSIM provisioning (ACM WiSec 2025):** Built the testbed, measurement pipeline, and analysis tooling that established the performance–consistency trade-off in the eSIM M2M remote provisioning protocol.

Industry Experience

Apple Inc. May 2025 – Aug 2025
Software Engineering Intern, Accessibility *Cupertino, CA*

- Implemented an iOS feature end-to-end (design, implementation, testing) in **Swift** within the Accessibility framework stack; shipped to internal beta after cross-team design and code review.

Agoda (Booking Holdings) Jan 2022 – Jun 2023
Senior Software Engineer *Bangkok, Thailand*

- Increased checkout conversion by designing and shipping a modular multi-product cart and checkout module for the Agoda iOS app.
- Led migration of the app's analytics layer from shared C# to native **Swift**.

Publications & Disclosures

G M T. Alam, Omar Chowdhury. *Implementation Aware 5G Core Fuzzer.* Under submission, 2026.

M Echeverria, A Sadeghi, **G M T. Alam**, Omar Chowdhury. *Performance and Consistency Trade-off of the eSIM M2M Remote Provisioning Protocol.* ACM WiSec 2025. [DOI]

26 CVEs in open-source 5G cores via coordinated disclosure, including CVE-2026-30075 (Critical, CVSS 9.8). Full list: shovon0203.github.io.

Other Projects

Supply-Chain Security Telemetry for CI/CD | *osquery, Kafka, Elasticsearch, Docker* 2024

- Built a streaming pipeline (osquery → Kafka → Elasticsearch) collecting kernel- and workflow-level events from GitHub Actions runners to detect compromised build steps.

Malicious Repository Detection at Scale | *Python, ELK stack* 2024

- Implemented scalable pipelines and detection heuristics to flag likely-malicious repositories across 100k open-source repositories.

NYC Motor Vehicle Collision Analysis and Visualization | *Python, D3.js* 2024

- Developed an interactive **D3.js** visualization pipeline over large public collision datasets to surface collision trends, hotspots, and risk patterns

Technical Skills

Languages: C, C++, Python, Swift, Java
Systems & Tools: Linux, Git, Docker, Kafka, Elasticsearch, osquery, GitHub Actions (CI/CD), Xcode
Research Areas: Fuzzing & automated testing, program analysis, crash triage & root-cause analysis, protocol security (5G NGAP/NAS, X.509/PKI, eSIM), experimental design & measurement

Service, Teaching & Awards

Cyber POWDER Fellow, University of Utah (2025) · **Reviewer**, IEEE Transactions on Dependable and Secure Computing

Teaching Assistant, Stony Brook (4 courses): Software Engineering, Systems Fundamentals, Game Programming, Logic in CS